Chain Pulse

- **PROBLEM STATEMENT** :Real-Time On-Chain Event Indexer & Notification System
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- **TEAM MEMBERS:**

MEMBER 1- Aryan Dangat 19

MEMBER 2 - Spandan Deb 20

MEMBER 3 - Sanket More 35

MEMBER 4 - Atharva Prabhu 45

PROPOSED SOLUTION

PROBLEM STATEMENT: RAW DATA, MISSED SIGNALS

Although publicly available, monitoring meaningful **on-chain** events in real time is fragmented and non-scalable. Teams rely on block explorers or basic listeners that expose **raw** transactions without **context** or **reliability**, resulting in noisy alerts, missed signals, and slow decisions—highlighting the need for a system that delivers **timely, reliable, and actionable insights**

PROPOSED SOLUTION: THE GAME-CHANGER

- **Unified Intelligence:** Query any contract or event from one rule-based interface.
- **Signal-First:** Filters blockchain noise into context-aware, actionable alerts.
- **Reorg-Safe:** Finality-aware architecture ensures accuracy in volatile conditions.
- **Instant Delivery:** Real-time webhooks, Telegram, or email notifications.
- **Lean Scaling:** Selective indexing slashes costs while maintaining performance.

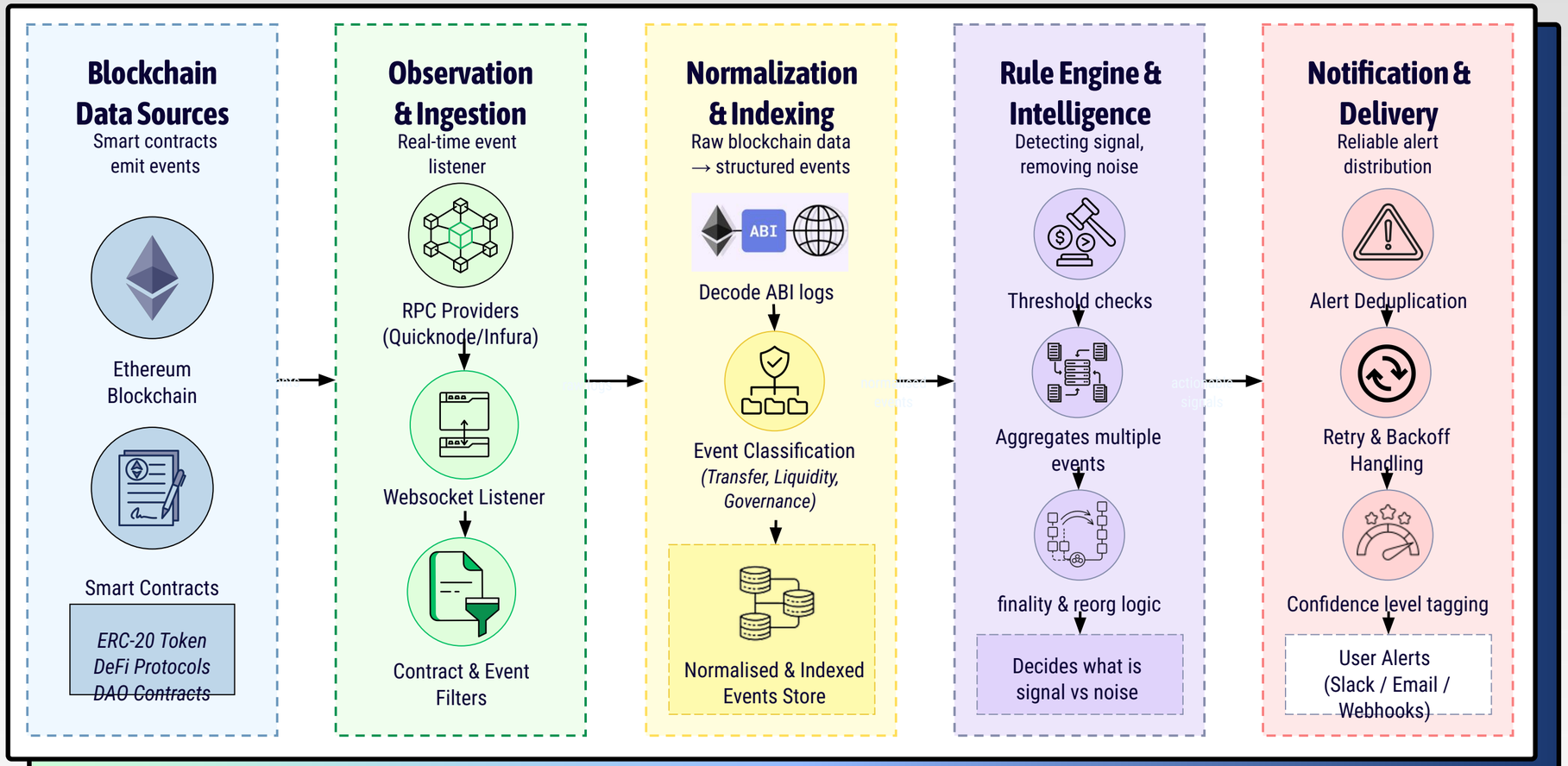
UNIQUE VALUE PROPOSITION

- **Finality-Aware Alerts**, not instant but unreliable notifications
- **Reorg-Safe by Design**, avoiding false positives
- **Index What Matters**, not the entire chain
- **Insights over Events**, optimized for human decisions
- **Built for Real-World Web3**, not demos

PROBLEM TO SOLUTION: Why Existing Monitoring Fails — And How We Fix It

Problem	Existing System (The "Why it Fails")	Our Solution
On-Chain Noise	Surfacing raw events without context or prioritization.	Signal-first event filtering.
Data Reliability	Vulnerable to reorgs and confirmation uncertainty.	Finality-aware processing.
Alert Fatigue	Manual filtering of high-volume, low-value data.	Smart event aggregation.
Lack of Context	No explainability; difficult to verify data sources.	Deterministic, auditable rules.
Operational Delay	Manual, time-consuming verification/monitoring.	Automated real-time alerts.

TECHNICAL APPROACH



TECH STACK

Blockchain

Ethereum

Data Access

Quicknode

Infura

Frontend

Next.js

Backend

Node.js

Typescript

ether.js

Data & Infra

PostgreSQL

Redis

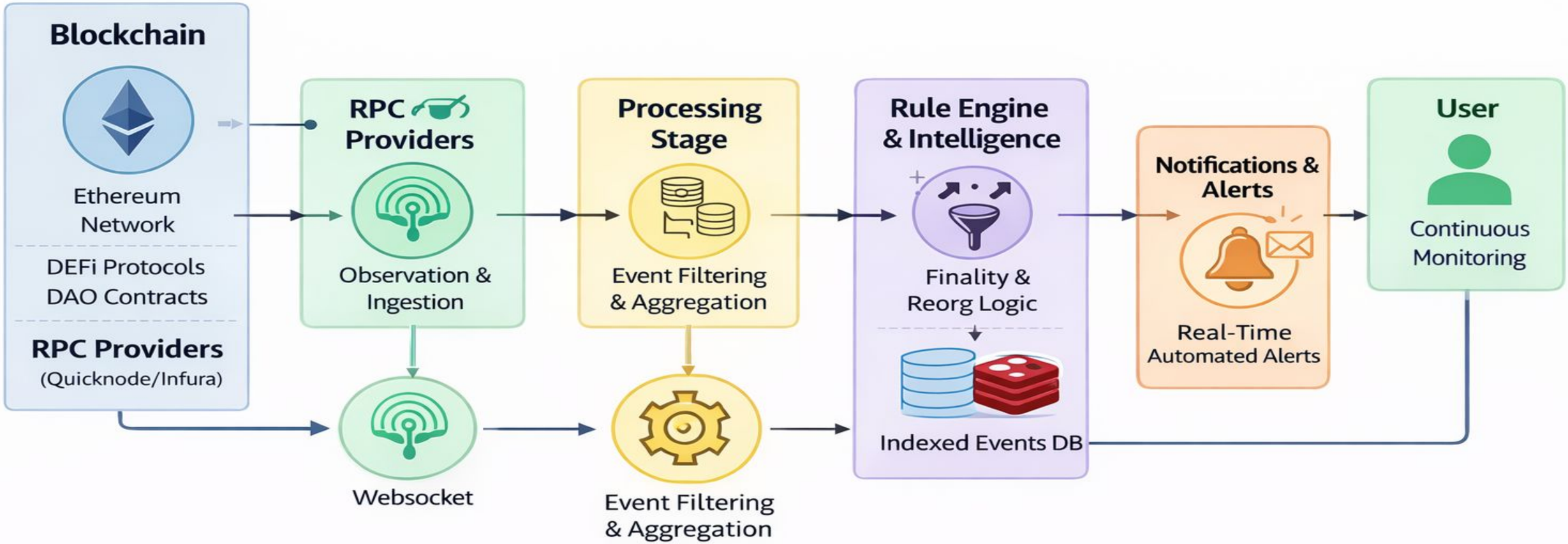
Delivery

webhooks

SlackAPI

- Ethereum → Most mature smart-contract ecosystem; trusted on-chain event source
- QuickNode + Infura → Low-latency streams with high availability & redundancy
- Node.js + TypeScript → Fast, safe, event-driven backend for real-time processing
- ethers.js → Reliable Ethereum contract interaction & ABI decoding
- PostgreSQL → ACID-safe, query-efficient storage for indexed on-chain data
- Redis → Scalable, fault-tolerant async event handling
- Webhooks + Slack API → Instant delivery to apps and ops teams

SYSTEM ARCHITECTURE



IMPACT AND BENEFITS

Real-world use case:

- **DeFi Lending Platform User Alert System**

This system can be used by a DeFi lending platform to notify users in real time when their collateral value drops, liquidation risk increases, or loans are repaid or liquidated, based on live on-chain events. It helps users take timely action and prevents unexpected financial losses.

IMPACTS

- **Reduced Signal Blindness in Web3:** Converts overwhelming on-chain activity into meaningful, prioritized signals for users and teams.
- **Faster, Confident Decision-Making:** Enables real-time responses to critical events like large transfers, protocol risks, and governance actions.
- **More Reliable Web3 Infrastructure:** Finality-aware and reorg-safe alerts reduce false signals and increase trust in blockchain monitoring systems.

Efficiency

Users react to token moves, unlocks, and governance events instantly.
Eliminates manual dashboard refreshes and polling.

Feasibility

Uses public chain data and standard Web3 tooling.
Easy to extend to new chains and contracts by adding listeners, not redesigning systems.

Security

Leveraging immutable on-chain logs with strictly read-only access.
Zero asset custody ensures no write-risk or vulnerability to user funds.

Scalability

Horizontal scaling via stateless listeners and workload sharding (by chain/contract) ensures high performance as event volume grows.

Cost Effectiveness

Selective indexing and finality-aware processing slash RPC calls, compute, and storage, driving down monitoring costs.

Reliability

Reorg-safe and finality-weighted alerts ensure users act on reliable information, minimizing false positives and operational risk.

COMPARISON WITH EXISTING TOOLS

Why Chain Pulse is Better

Feature	Block Explorers	The Graph	Traditional Listeners
Real-time Alerts	Manual Only	No	Custom Dev
Context & Decoding	Raw Logs	Limited	JSON payload
Reorg Handling	None	Partial	Manual
Noise Filtering	No	Basic	Custom
Setup Effort	None	Complex	High
Best For:	Viewing Data	Quetying	Developers

Bottom Line: Chain Pulse – Real-time, Actionable.

SECURITY CONSIDERATIONS

Built with Security First Mindset

Read-Only Access
- No wallet access
- No fund control

No Private Keys
- No custody
- Only public data

Secure APIs
- Encrypted endpoints
- Rate limiting

Audit Trail
- Immutable logs
- Full transparency

Your Funds Are Never at Risk. We Only Read Public Data.

RESEARCH AND REFERENCES

Future Enhancements

- **Multi-Chain Support**: Extend beyond Ethereum to Polygon, Arbitrum, Solana
- **ML Signal Ranking**: Anomaly detection + priority scoring for critical alerts
- **Natural Language Rules**: Users define alerts like "Notify if >$1M whale move"
- **Enterprise Integrations**: PagerDuty, Discord, SIEM, governance dashboards
- **Scalable Deployment**: Distributed workers, sharded indexing, high-throughput pipelines
- **Security Expansion**: Stronger audit trails, role-based access, proof anchoring

Comparative Analysis

Features	Traditional Tools	Our System
Meaningful Signal Detection	✗	✓
Finality-Aware Alerts	✗	✓
Reorg Handling	✗	✓
Aggregated Insights	✗	✓
Real-Time Notifications	!	✓
Selective Indexing	✗	✓
Production Reliability	!	✓

Target Audience

aave

UNISWAP

DeFi Protocol Teams

DAO & Governance Operators

Our product aims to become the trusted real-time intelligence layer for on-chain activity.

TOOLS & REFERENCES

- **Ethereum Event Logs + ABI Decoding** (ethers.js)
- **QuickNode / Infura RPC Providers for low-latency ingestion**
- **Node.js + TypeScript Backend with PostgreSQL + Redis**
- **Verified Alerts & Finality Concepts** (Ethereum Documentation)
- **Delivery Channels**: Telegram, Slack, Email, Webhooks

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REAL-WORLD USE CASES

- DeFi exploit & liquidity drain early-warning
- DAO treasury monitoring & governance oversight
- Whale tracking for exchanges and institutions
- Smart contract upgrade + admin risk detection

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